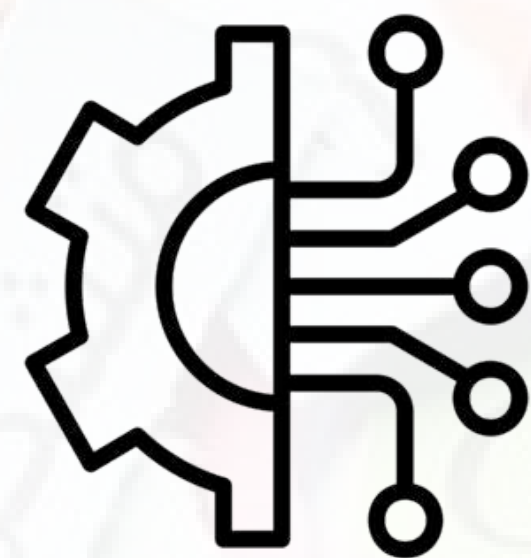




LEUVEN STATISTICS
RESEARCH CENTRE



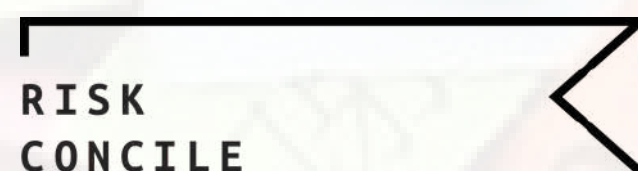
KU Leuven

DATATHON 2026

KICK-OFF



Made possible by



Deloitte.

WELCOME STUDENTS!

We are thrilled to welcome all of our talented and motivated students who have accepted the challenge to torture the data until it confesses

Without students,
no **Datathon @ KU Leuven!**

KU LEUVEN

LEUVEN STATISTICS
RESEARCH CENTRE





WELCOME SPONSORS!

We are equally excited to welcome
our **generous sponsors!**

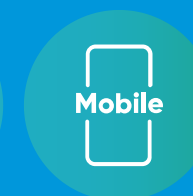
Without sponsors, no datathon @
KULeuven! (or it would be one
without food, drinks and prizes...).
Take this moment to introduce
yourselves.

Deloitte.

RISK
CONCILE



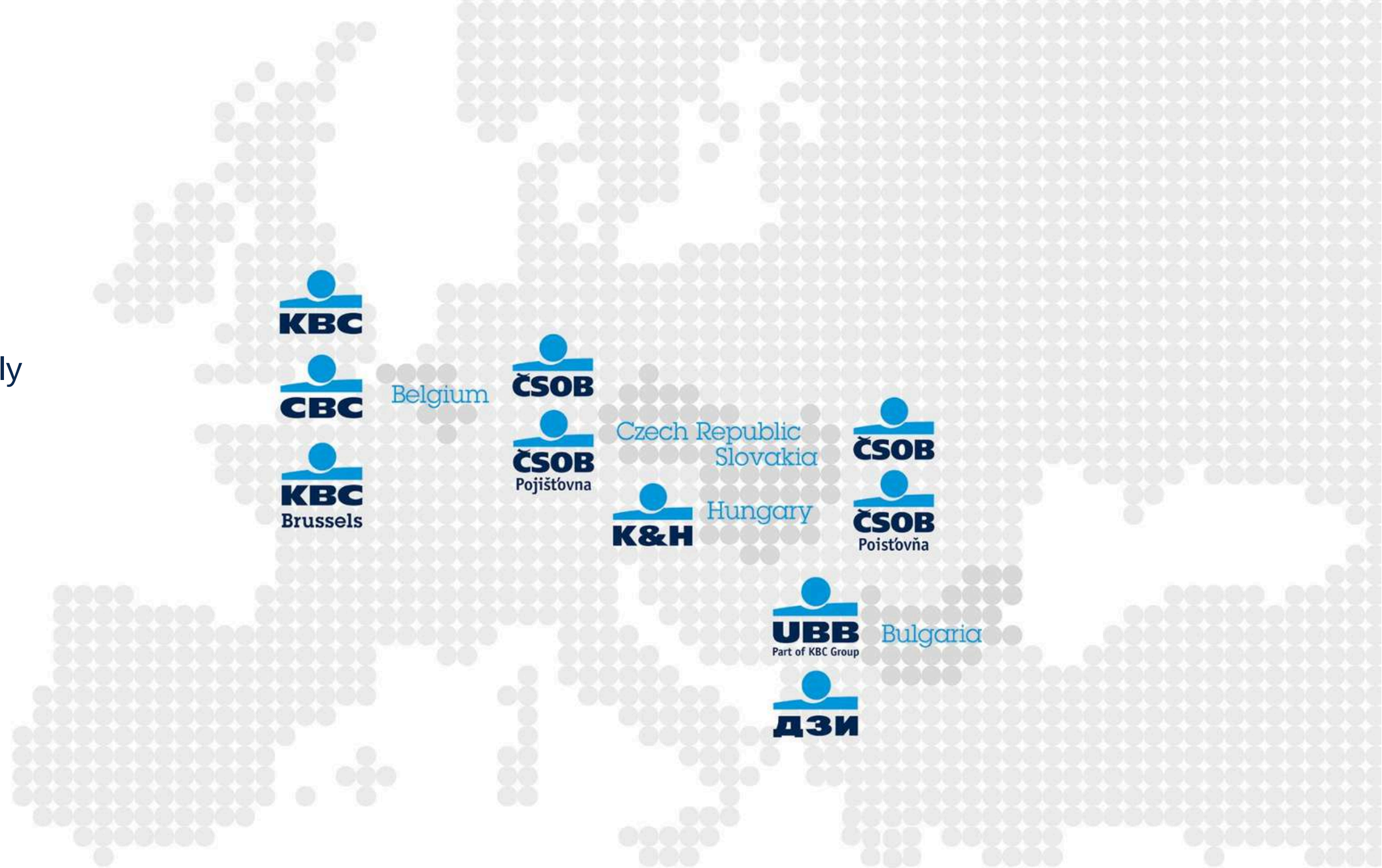
KBC @ KU Leuven Datathon 2026



Moving forward
together.

KBC Group

- Integrated bank-insurance group, catering mainly for retail, private banking, SME and mid-cap clients.
- Core markets – Belgium, the Czech Republic, Slovakia, Hungary and Bulgaria. Limited presence in several other countries.
- Formed in 1998 after the merger of two Belgian banks (Kredietbank and CERA Bank) and a Belgian insurance company (ABB Insurance).
- Clients: 13 million
- Employees: 40 000
- Bank branches: 1090





Driven by data
Powered by *ADAM*

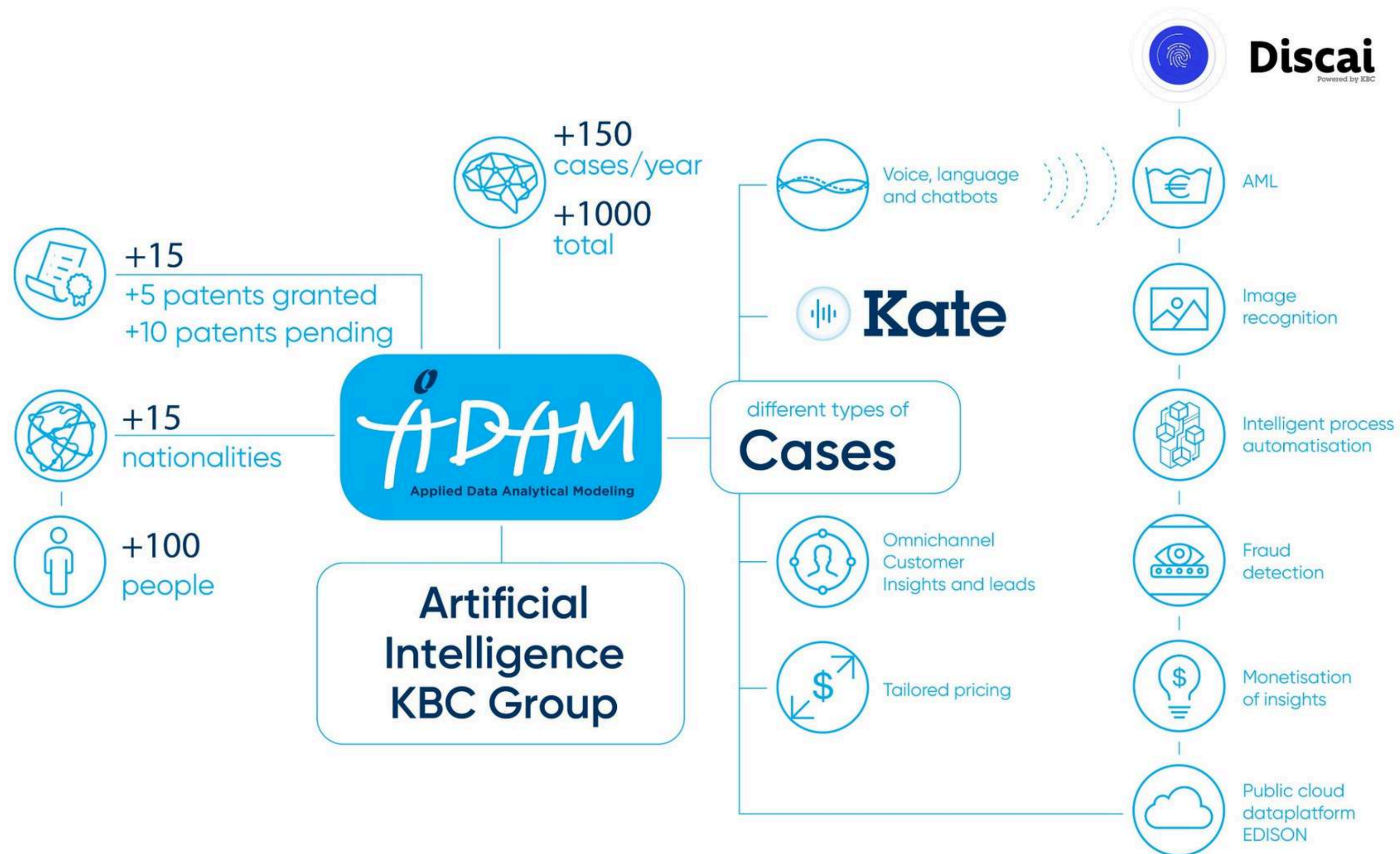


Team Blue

“Data is the new
electricity”

Johan Thijs, CEO KBC Group





Multidisciplinary and multicultural team



ADAM BE Team

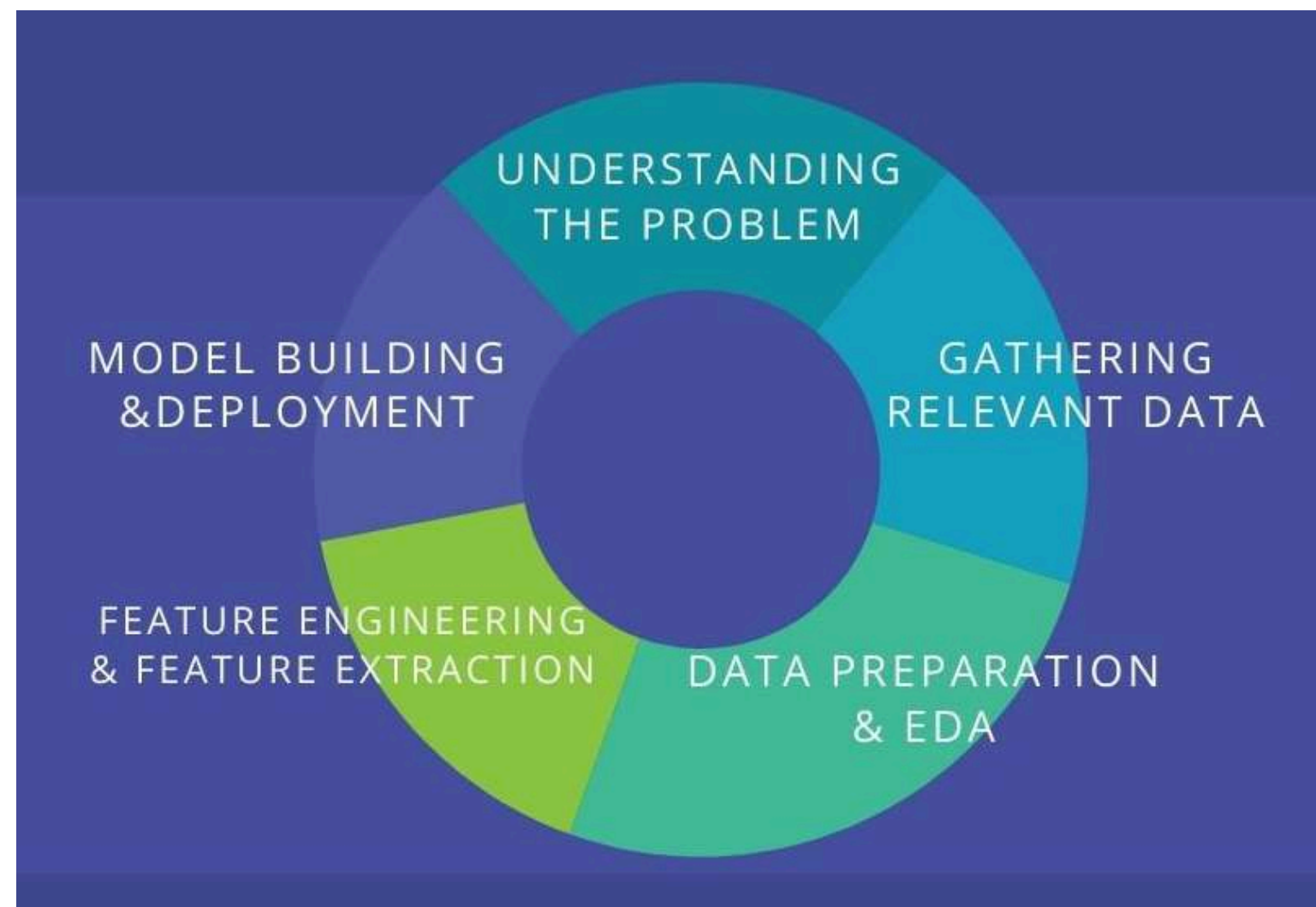


ADAM Bootcamp 2025 (Bulgaria)



ADAM BE Team

Being a Data Scientist in ADAM



Full data science project lifecycle

A data scientist takes a **business challenge** and translates that into an **AI solution** using the variety of data we have in KBC. You don't just sit behind your computer but have plenty of **interactions** and inspiring **discussions** with your colleagues and stakeholders.

What you build brings **real value** to the company

You **stay up-to-date** with cutting edge techniques

You work on a **variety** of problems

Expertise domains in ADAM

Pricing & Client
valuation

Conversational AI
(Kate)

(Gen) AI assistance
for employees &
Agents

Financial Markets
& Economics

Anti-money
Laundering

Real Estate

Document
processing

Sales

Lead
management &
Kate





Kate

Kate is best Bot of Belgium 2025



1st Place



2nd Place



3rd Place



Kate

KBC

Total score

48.5/50

Bot Intelligence

19.5/20

Conversation Design

15/15

Out-of-scope Management

5/5

Visualization

5/5

Accessibility

4/5



KBC Mobile is once again crowned best banking app in the world!
2021 – 2024 -2025

Your KBC Datathon Team



Deepak Baby

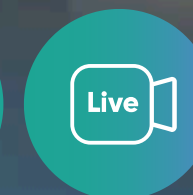


Emmanouil Mavrogeorgis



Mark Casselman

Don't be shy.



Moving forward
together.



Deloitte Belgium is the leader in Digital Technology and Business consulting
Some numbers for 2025:

Headcount: 5000+
New Hires: 1500
Revenue: 822 Mio Euro
Net Revenue Growth: +1%

Choose your impact!

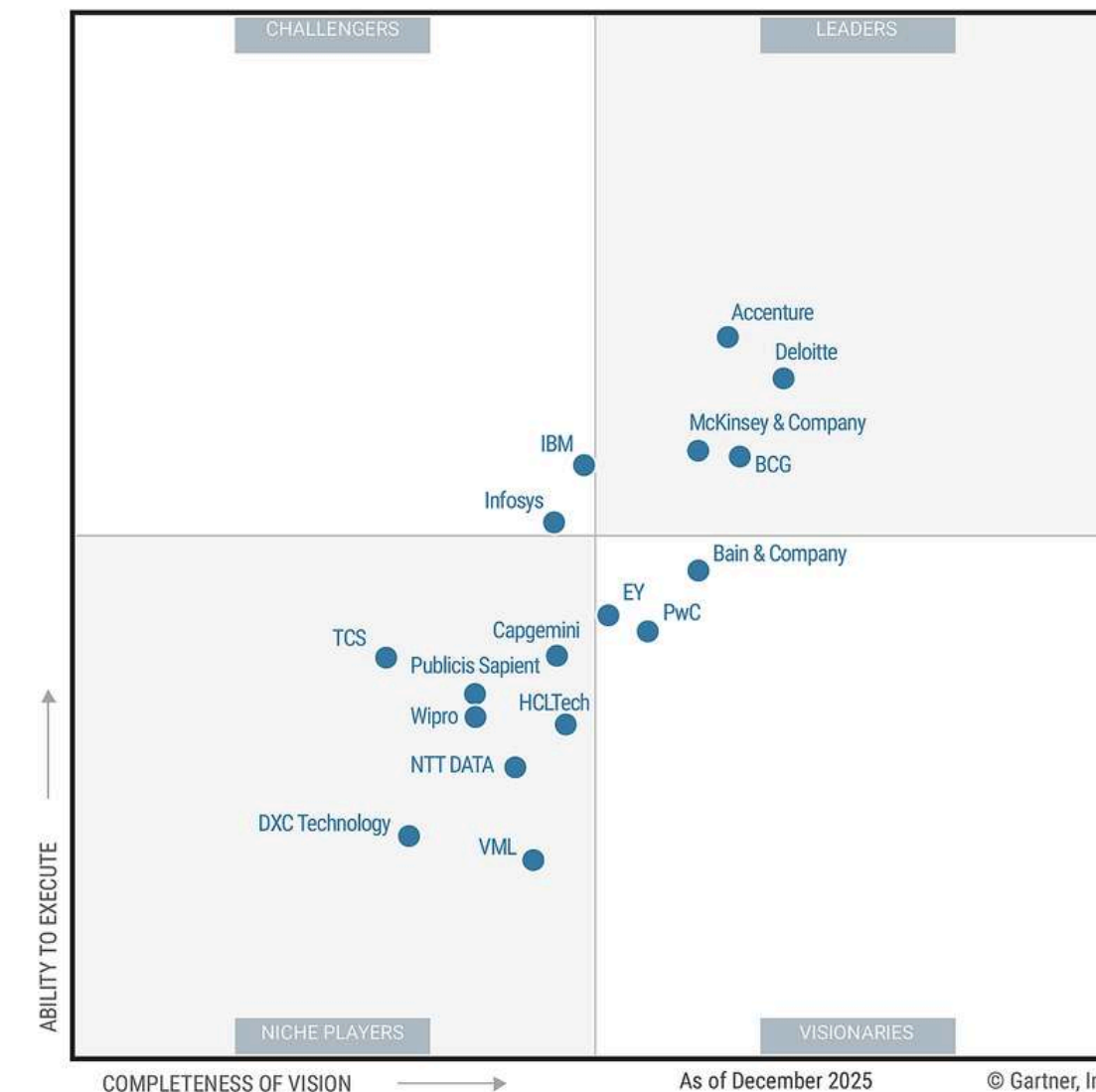
deloitte.com/be/mycareer

Enterprise Technology and performance

Cyber

Human capital

Figure 1: Magic Quadrant for Digital Technology and Business Consulting Services



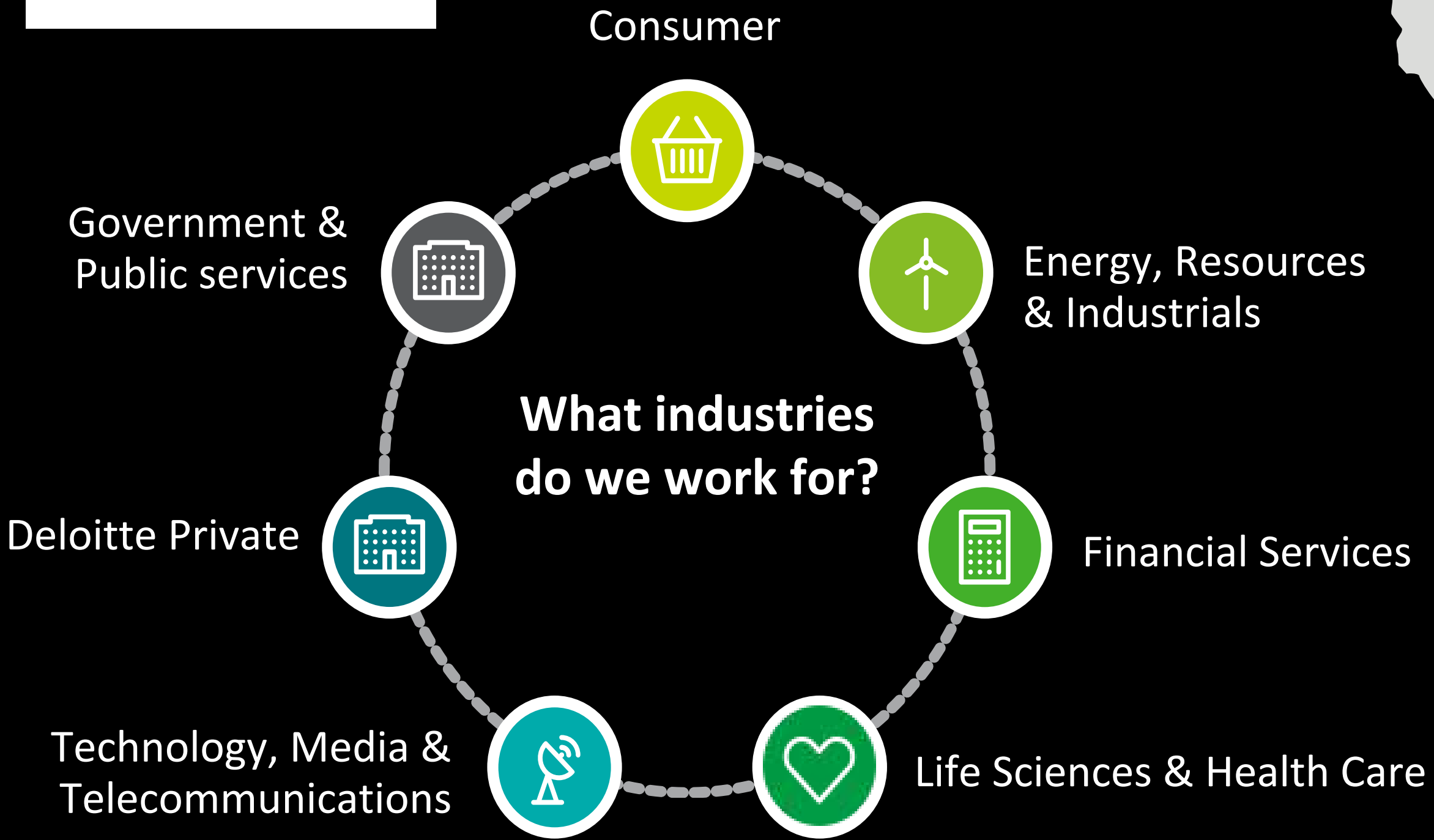
Strategy and Transactions

Finance Transformation

Engineering AI and data



Let's connect!
You can apply on our career website





Meet our people



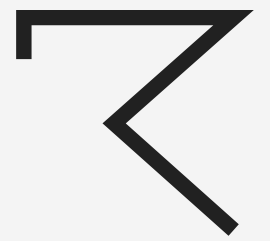
Kris Bornauw
Senior Director
Enterprise architect
Innovation lead



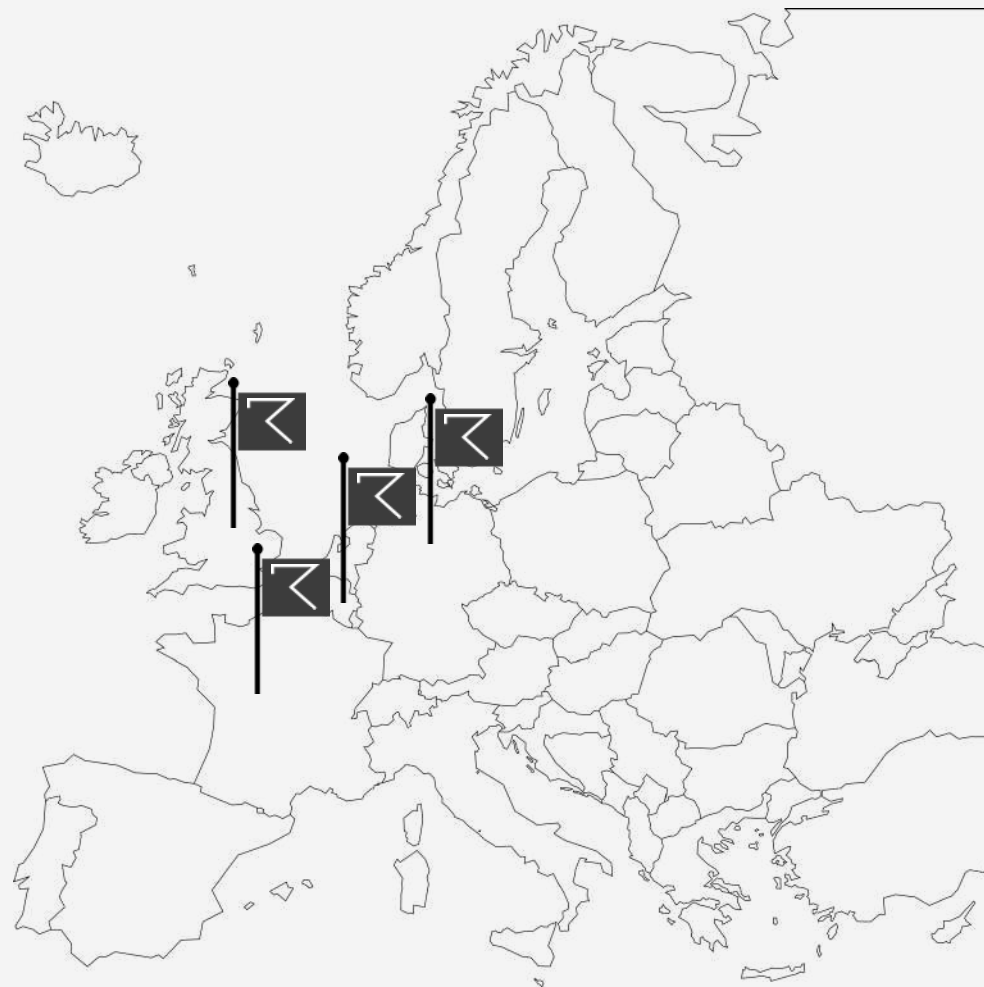
Allan De Brueker
Director
Architecture and Technology offering
leader

RiskConcile Group

19 February 2026



RiskConcile Group



Founded in **2015**, RiskConcile Group provides solutions to the **asset management** industry

c. 90 people across multiple offices in Europe and particular strong exposure to **UK, Ireland, France, Luxemburg** and **DACH**

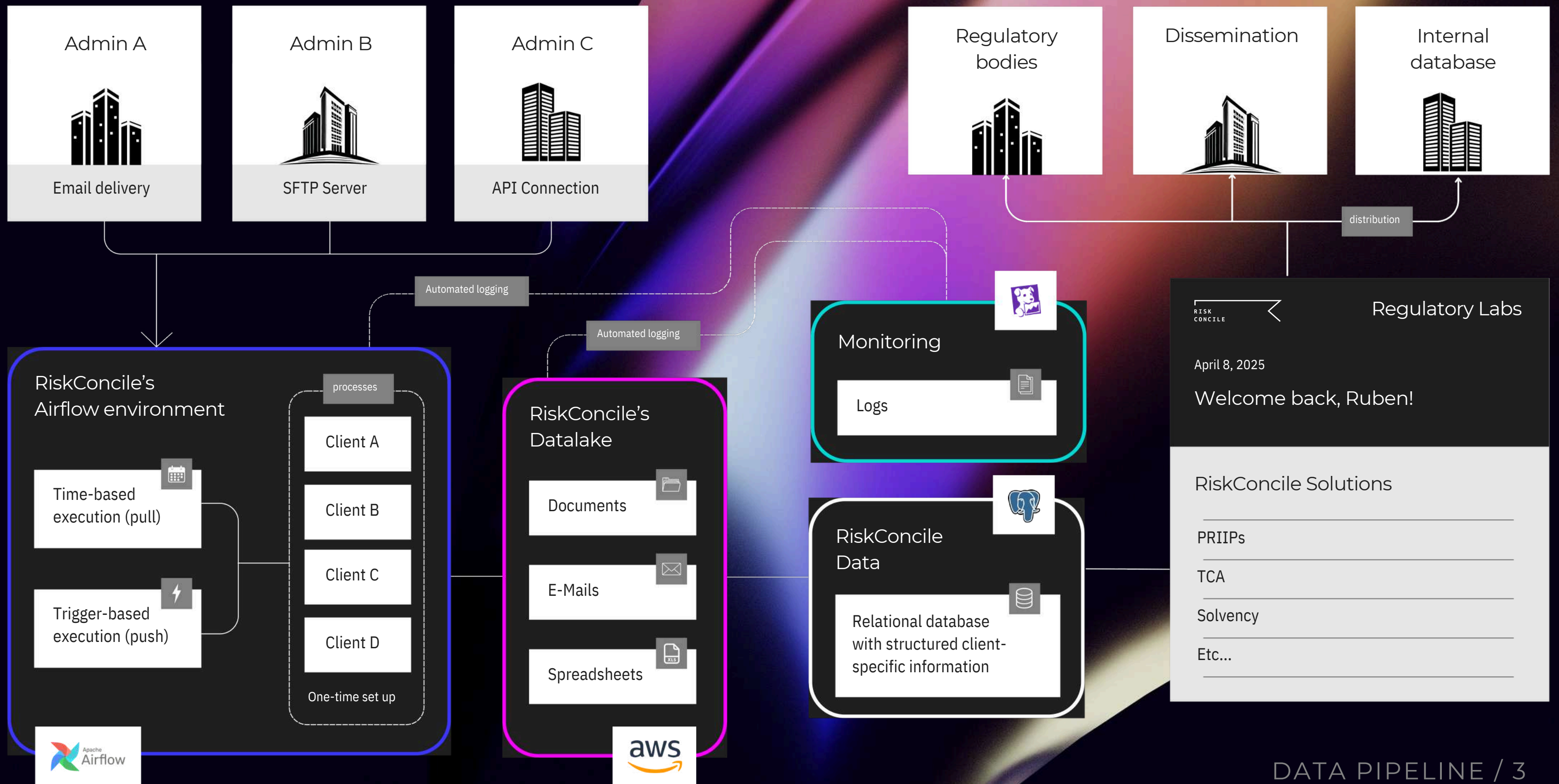
3 complementary businesses units:

- Risk Calculation and Regulatory Reporting
- Fund fee data (Assessment of Value Benchmarking)
- KYC/AML due diligence (all counterparties)

Deep expertise in both Public Markets as Private Markets



“A pan-European provider of Risk & Regulatory Cloud-Based Software Solutions for the asset management industry. Recognised for its thought leadership supported by in-house software development team and strong data-discipline.”



OUR TECH STACK

TECHNICAL SETUP



APPLICATION ENVIRONMENTS



ECS



S3



Lambda



RDS



Batch



SQS

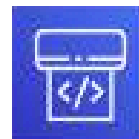
DEVOPS



git



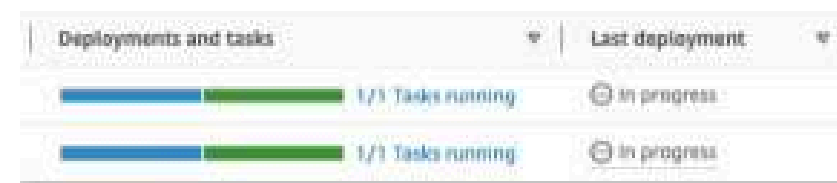
docker



CodePipeline



CloudFormation



Zero-downtime deployments

MONITORING



Application Logging



DATADOG



CloudWatch



Alternative Tech Stack



Your Partner in a regulated World

The Data

Background information
and key features



Translate this sentence

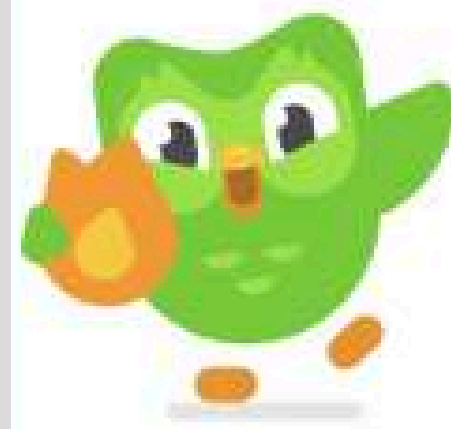
All data sets this year were made available by **Duolingo Research**

What is Duolingo?

- Duolingo is a **learning platform**, mainly focused on teaching **languages** (also math, music, ...)
- Courses in **42 languages**
- **130 million** monthly active users (2025)
- **Gamification** for motivation

Commit to learning Spanish!

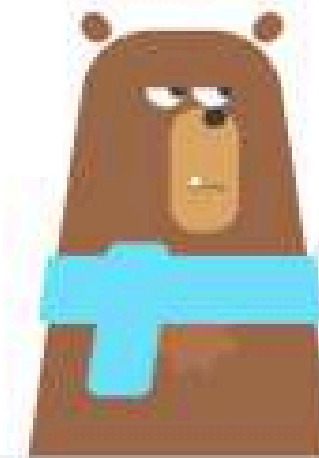
7 day streak	Good
14 day streak	Great
30 day streak	Incredible
50 day streak	Unstoppable



You'll be **2x more** likely to complete the course!

COMMIT TO MY GOAL

Repeat what you hear



REVEAL



TAP TO SPEAK

CAN'T SPEAK NOW

CONTINUE

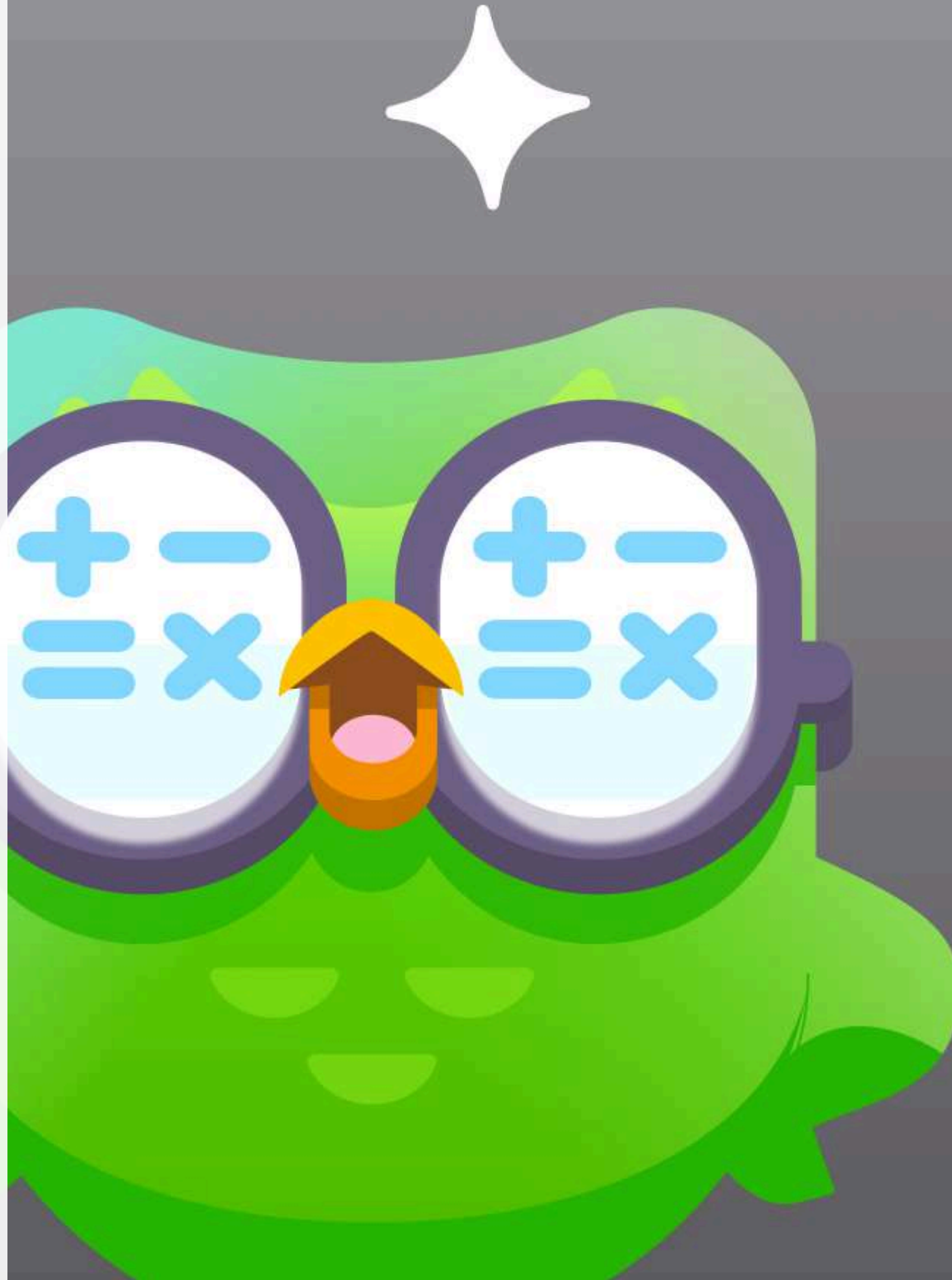
THE DATA SETS

This year, we have **THREE** data sets

Each data set covers an important part of the Duolingo platform

- **User Engagement:** How to motivate people to practice?
- **Quality of Education:** How to ensure users get the most out of their efforts?
- **Quality of Content:** How to ensure the exercises are correct?

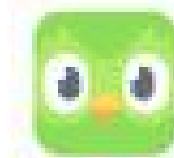
The **data sets** of this year were constructed to **answer** these **questions**



THE DATA SETS

User Engagement:

- The key to learning languages is **regular practice**
- In 2021, less than 10% of users lasted a week!
- Duolingo uses **notifications** to remind/motivate users to practice
- But **what** should the notification say? And **when** should they send it?

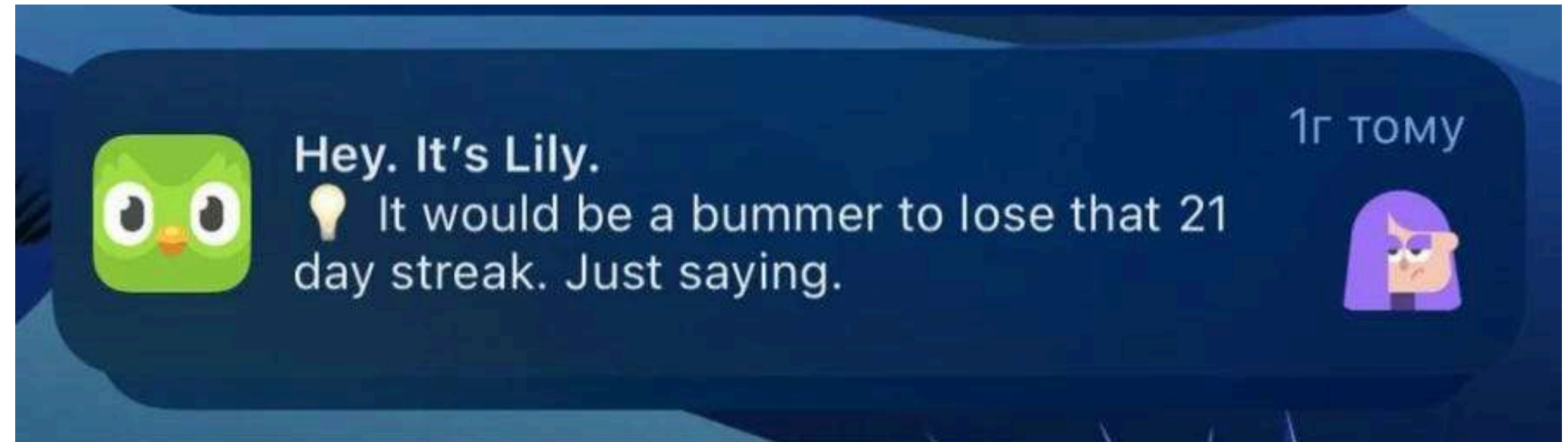


new notification

did you do
your duolingo
today?

THE DATA SETS

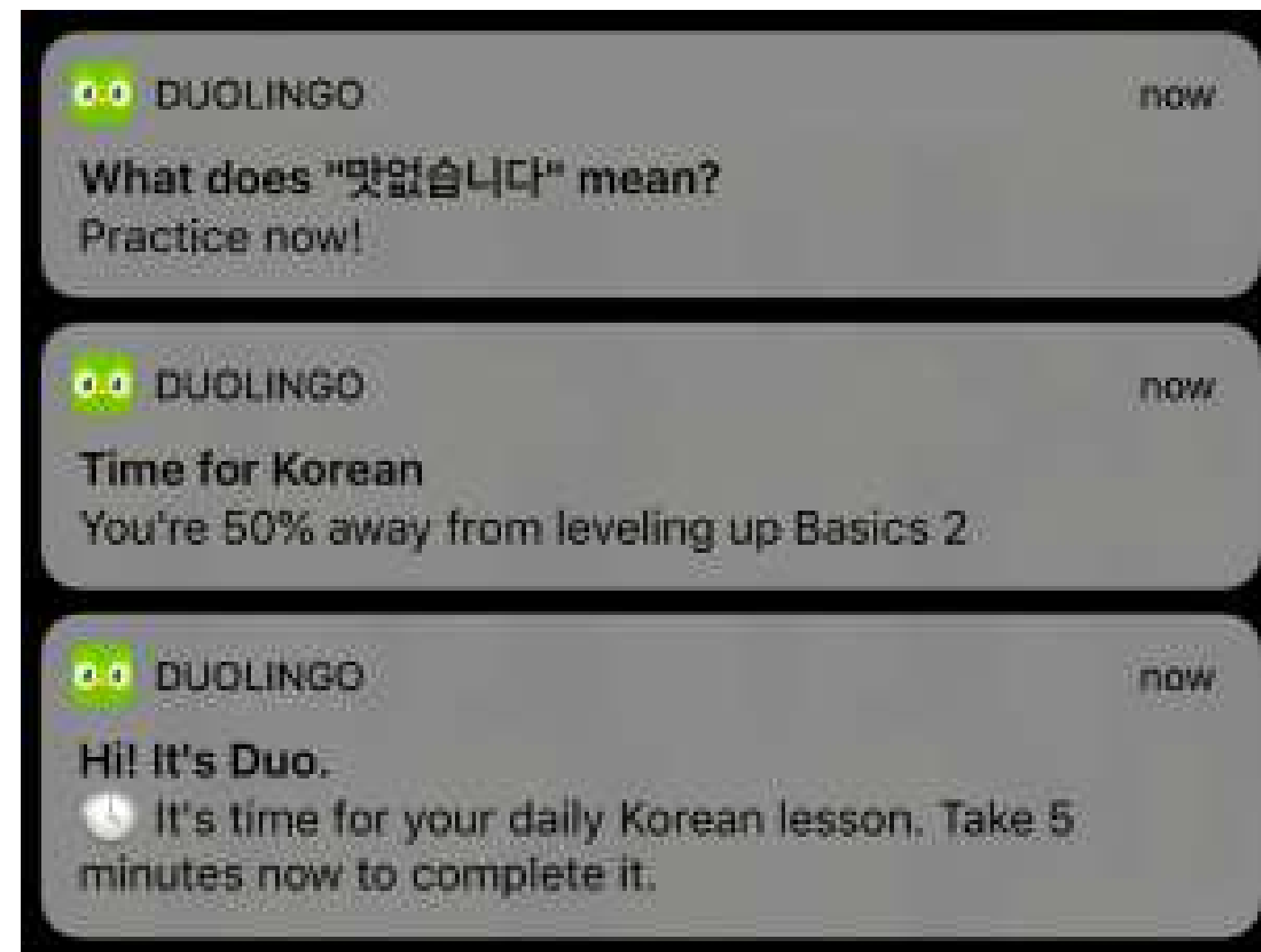
User Engagement:



2020 Notification Bandit

- The Duolingo Research team uses **machine learning** to **optimize** the notification system
- Build a massive data set containing information on more than **200 million notifications**
- Informative features:

Timestamp, UI language, eligible templates, historical performance, ...



THE DATA SETS

Quality of Education:

- **Revision** is important...
- ... but how does Duolingo decide **which** words to **revisit**?
- Duolingo uses a **personalized review schedule** based on **spaced repetition**
- Their findings have been published in a **paper** (also included in the Google drive!)



THE DATA SETS

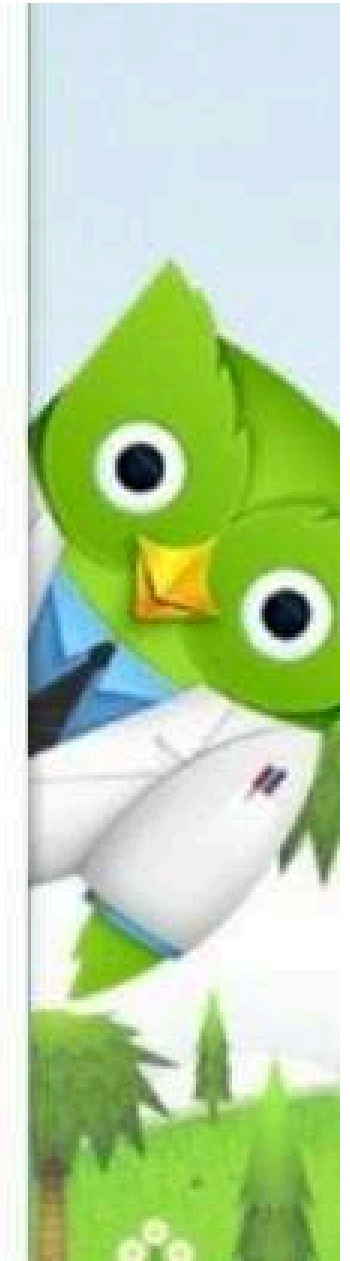
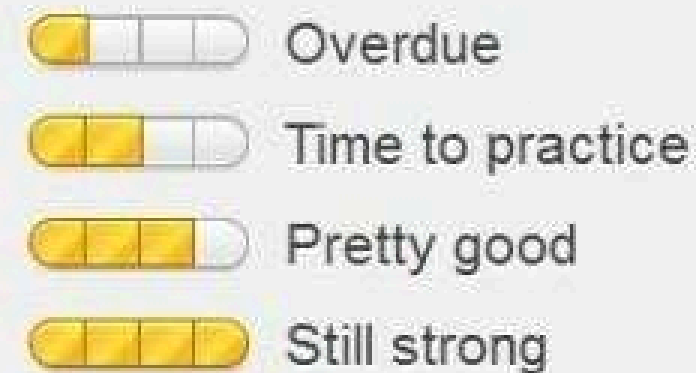
Quality of Education:

Spaced Repetition:

- Cramming is not good for **long-term memory retention**
- Spaced Repetition is a method where you review information **just before it is forgotten**
- **Newer words** are tested more frequently
- How do we apply this theory in practice?

Spaced repetition

Duolingo's algorithms figure out when you should practice words to get them into your long-term memory.



THE DATA SETS

Quality of Education:

- Duolingo Research released a paper introducing their **Half-Life Regression model**
- Using this model, they generate **personalized review schedules** for their users
- The model is built on a large data set consisting of 13 million records

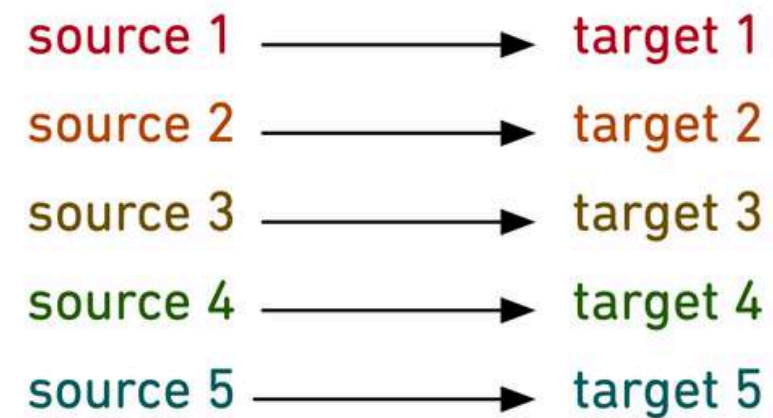
Features: History for each word, total exposure, total correct recalls, word difficulty



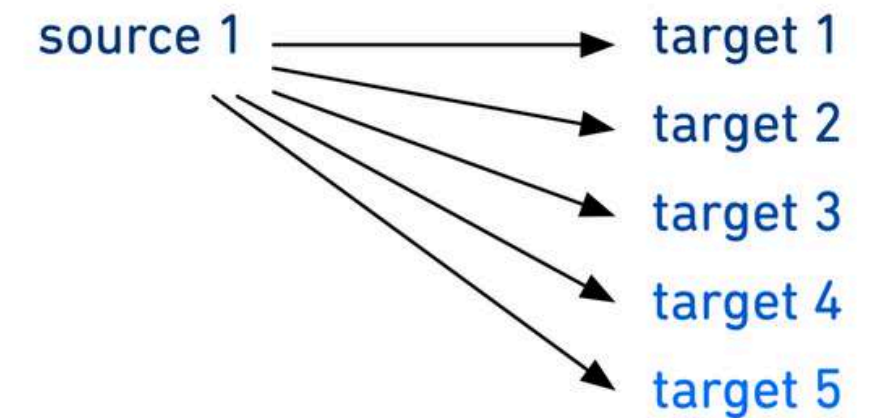
THE DATA SETS

Quality of Content

- Duolingo uses **Machine Learning** to translate sentences (e.g., in exercises)
- These models often map each sentence in the source language to a **unique** sentence in the target language
- In reality, **multiple** correct translations exist
- Important when grading users!



Generic Translation Data



Duolingo STAPLE Data

THE DATA SETS

Quality of Content

- This **human-curated** data set consists of 5 **language pairs**
- Each observation consists of a sentence in English, together with a **list** of possible translations with **scores**
- The scoring is based on how often a specific translation is used by **real users**
- This data set has been made available by the Duolingo team
- Part of a competition in 2020 hosted by Duolingo

en_pt	— Portuguese
en_hu	— Hungarian
en_ja	— Japanese
en_ko	— Korean
en_vi	— Vietnamese

INPUT: Prompt	is my explanation clear?
INPUT: Reference Translation (from Amazon)	a minha explicação está clara?
OUTPUTS: Accepted Translations	minha explicação está clara? minha explicação é clara? a minha explicação está clara? a minha explicação é clara? minha explanação está clara? está clara minha explicação? minha explanação é clara? a minha explanação está clara? é clara minha explicação? a minha explanação é clara? está clara a minha explicação? é clara a minha explicação? está clara minha explanação? é clara minha explanação? está clara a minha explanação? é clara a minha explanação

THE DATA SETS

General Remarks

- You are allowed to use **relevant data** you find online
- You are free to choose
 - What **tools** you use
 - How you **present** your work
 - What **research question** you answer

THE CHOICE IS YOURS

- Read the README files in the drive for additional context!



When?

Wanneer?

Когда?

Коли?

什么时候?

いつ?

متى؟

Ne zaman?

कब?

几时?

Kailan?

¿Cuándo?

Quand?

Ntango
nini?

Quando?

Lini?

Kapan?

LET'S GET PRACTICAL

February 2026

Mon

Tue

Wed

5

MARK YOUR CALENDARS

February 26

14h00 - DEADLINE

• Who?

Team captain

• Where?

Google Drive (Shared in due time)

• What?

Solution (can be in any format),
no data

The earlier, the better!!

15h45 - Voucher Pick-Up

16h15 - Start Pitching

19h00 - Award Ceremony

19h30-21h00 - Networking

9

23

24

25

10

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Today

Thu

Fri

Sat

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7

February 26

Sat

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7

14

21

28

February 2026

Mon

Tue

Wed

MARK YOUR CALENDARS

February 26

14h00 - DEADLINE

15h45 - Voucher Pick-Up

16h15 - Start Pitching

- Teams **pitching**
- Team captain will be informed beforehand where and when.

Ground rules:

- Use your **own laptop**.
- There are **HDMI** cables (but no HDMI Thunderbolt adaptors)
- Be at least **5 minutes earlier**
- Pitching **behind closed doors**
- Time limited to **5 minutes**
- Jury Q&A of 5 minutes after pitch
- Teams are graded based on modeling, pitching, and visualization.

19h00 - Award Ceremony

19h30-21h00 - Networking

<Today

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February 2026

Mon

Tue

Wed

MARK YOUR CALENDARS

February 26

14h00 - DEADLINE

15h45 - Voucher Pick-Up

16h15 - Start Pitching

19h00 - Award Ceremony

• Each member of a winning team receives a CoolBlue voucher

• Price categories:

16



Best Pitch

17



Best Visuals

18



Best Model

• Pitch in front of audience that votes for 'Winner of the Datathon 2026'

19h30-21h00 - Networking

< Today

Thu

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Sat

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MARK YOUR CALENDARS

February 26

14h00 - DEADLINE

15h45 - Voucher Pick-Up

16h15 - Start Pitching

19h00 - Award Ceremony

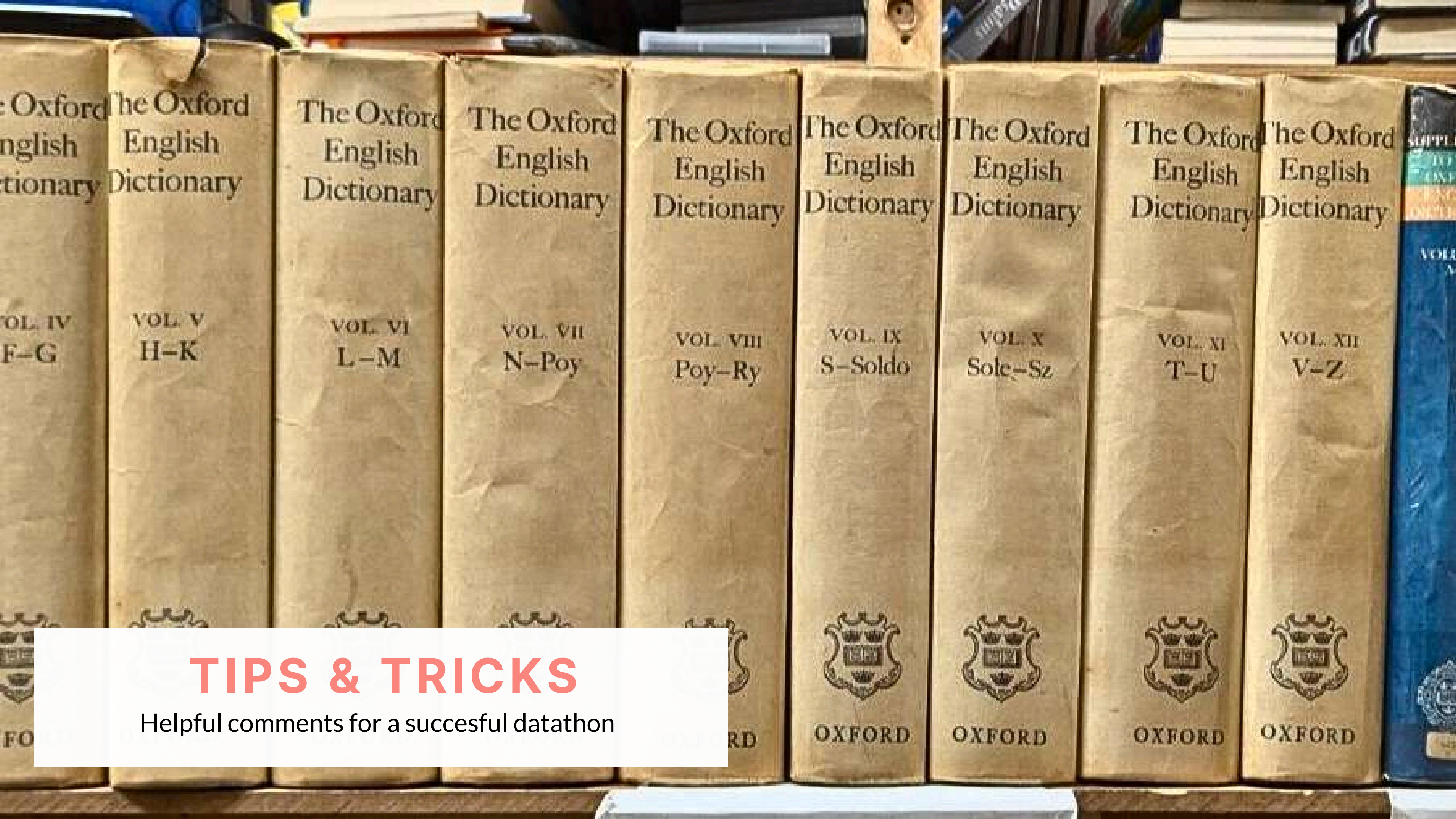
19h30-21h00 - Networking

- Have a well-deserved drink at the **closing reception** in **Quadrivium**
- Discuss your ideas and work with your peers
- **Network with our sponsors!**

RISK
CONCILE



Thu	Fri	Sat	Sun
29	30	31	1
5	6	7	8
12	13	14	15
19	20	21	22
26	27	28	1
5	6	7	8



The Oxford
English
Dictionary

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VOL. IV
F-G

VOL. V
H-K

VOL. VI
L-M

VOL. VII
N-Poy

VOL. VIII
Poy-Ry

VOL. IX
S-Soldo

VOL. X
Sole-Sz

VOL. XI
T-U

VOL. XII
V-Z

TIPS & TRICKS

Helpful comments for a succesful datathon

TIPS AND TRICKS

Tooling

Choose the correct tool for the job

Consider several programming languages and packages before starting your analysis. Using the 'usual' packages is not a bad idea, but might not give you an edge over the competition

Choose the platform you're going to work on wisely

Have fun!



TIPS AND TRICKS

Tooling

Choose the correct tool for the job

Choose the platform you're going to work on wisely

Working together on the same code is hard, find a solution that fits your whole team!

(e.g., Google colab for Jupyter notebooks. Helpful for their free (but limited) GPU computation services)

Have fun!



TIPS AND TRICKS

Tooling

Choose the correct tool for the job

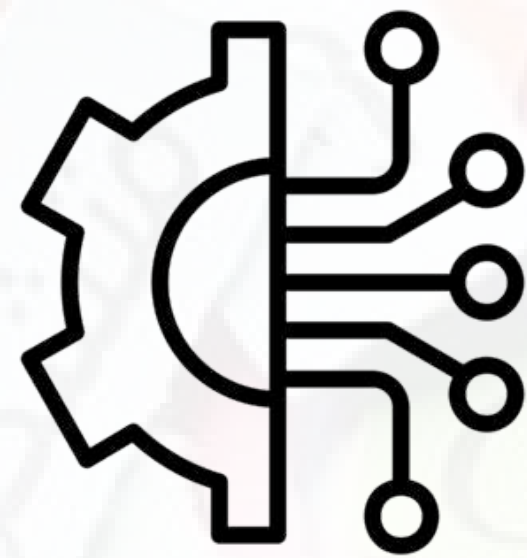
Choose the platform you're going to work on wisely

Have fun!

Remember that this is not graded! Communication is key!
... and most importantly, have fun!

Good Luck!





KU Leuven

WORKSHOP

AI in the Organization

The Agentic AI Opportunity: A Market Overview

From Simple Automation to Agentic AI

The manufacturing sector is at a pivotal moment, transitioning from simple automation to intelligent, agentic systems. While current automation provides incremental gains, the real opportunity lies in **agentic AI**, which promises holistic transformation and strategic value. This shift requires a strategic focus on scalability and ROI to unlock unprecedented efficiency and innovation. Success hinges on adopting an '**Understand Scale First**' approach to ensure that the move to agentic AI delivers sustainable, enterprise-wide impact.

Limited Adaptability

Current automation offers incremental improvements but often operates within rigid, pre-defined workflows, limiting its ability to adapt to **dynamic production** changes.

Isolated Efficiency

Simple automation focuses on isolated tasks, providing localized efficiency gains but failing to deliver **holistic business transformation** or unlock new innovation pathways.

Dynamic Operations

Agentic AI will drive a **400% growth** in adoption by enabling systems that autonomously manage complex processes and adapt to real-time manufacturing demands.

Strategic Value

This new paradigm moves beyond simple task execution to deliver **strategic value**, fundamentally reshaping how manufacturing operations are managed and optimized.

Strategic Imperatives



Understand Scale First

The market is on the cusp of a **transformative shift**, but relying on simple automation alone will not capture the full potential of this change.



Maximize ROI Potential

Agentic AI offers **fundamental business transformation** by creating intelligent, interconnected systems that unlock unprecedented levels of efficiency and drive continuous innovation.



Enterprise-Scale Deployment

Success requires an '**Understand Scale First**' approach to ensure the transition to **agentic AI** delivers transformative, sustainable value across the entire organization.

Deloitte's Trustworthy AI framework to protect your AI value & investments

TRADITIONAL AI CHATBOTS

REACTIVE RESPONSE

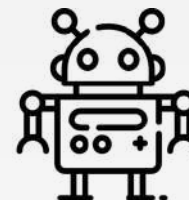
Traditional chatbots are reactive, responding to specific user prompts with information but lacking the ability to act beyond the immediate **information retrieval** task.

INTERACTION SCOPE

These systems operate within the confines of a **single interaction**, unable to retain context or learn from previous exchanges to inform future actions.

HUMAN DEPENDENCY

Progress is entirely dependent on **human direction** for each step, requiring constant manual input to guide the process and execute subsequent tasks.



Real-World Example: CNC Machine Maintenance

- A single prompt initiates an **autonomous workflow**.
- The agent provides instructions and notifies scheduling.
- It adjusts for **real-time conditions** from sensors.
- Finally, it logs completion data, closing the loop.

AGENTIC AI SYSTEMS

PROACTIVE AUTONOMY

Agentic AI systems operate with **agency**, proactively taking initiative and making decisions to achieve a goal with minimal human intervention.

ADVANCED CAPABILITIES

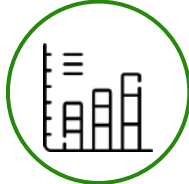
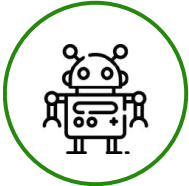
They possess persistent **memory** and access diverse **tool ecosystems**, allowing them to learn, adapt, and utilize external resources to accomplish complex tasks.

INDEPENDENT EXECUTION

These systems can **reason, plan, and execute** multi-step workflows independently, demonstrating a sophisticated level of autonomous problem-solving and task completion.

Business Value Across the Manufacturing Value Chain

Agentic AI delivers transformative value across the entire manufacturing ecosystem. From optimizing internal processes to revolutionizing production and enhancing customer interactions, its impact is comprehensive. This slide illustrates key value propositions in **Back Office Operations**, **Production & Manufacturing**, and **Customer-Facing Functions**, demonstrating the widespread benefits of adoption.

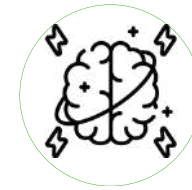
Back Office Operations	Production & Manufacturing	Customer-Facing Value
 TAX & FINANCE Automated data extraction streamlines tax and finance processes.	 QUALITY CONTROL Implement real-time quality inspection on the production line.	 SALES Accelerate sales cycles with automated quote generation.
 IT & CYBERSECURITY Enable real-time threat response to protect critical systems.	 EQUIPMENT REPAIR Enable autonomous identification of necessary equipment repairs.	 CUSTOMER EXPERIENCE Deliver proactive service to enhance customer satisfaction.
 HR/TALENT Utilize adaptive skills assessment for talent development.	 MAINTENANCE Optimize uptime with predictive maintenance scheduling.	 AFTERMARKET SERVICES Drive high-margin revenue growth with 2x margin potential.
 EFFICIENCY Enhance operational efficiency across all back-office functions.	 PRODUCTIVITY Boost shop floor productivity and reduce operational downtime.	 REVENUE GROWTH Unlock new revenue streams and improve customer loyalty.

Business Value Across the Manufacturing Value Chain



SUPPLY CHAIN RESILIENCE

Utilize autonomous agents for **disruption response**, alternate sourcing, and automated negotiation. This proactive approach significantly reduces supply chain risk while simultaneously optimizing operational costs, ensuring your business remains agile and competitive in a volatile market. The system intelligently adapts to unforeseen challenges, maintaining continuity and efficiency.



PRODUCT INNOVATION

Accelerate customer-centric design by integrating and analyzing IoT and service data. This holistic view of product performance and user interaction shortens the **time-to-market** for new innovations. By understanding customer needs more deeply, you can develop products that are not only innovative but also highly relevant and desirable, securing a stronger market position.



PRODUCTION EXCELLENCE

Achieve superior manufacturing outcomes through real-time quality optimization and **autonomous line reconfiguration**. This enhances operational agility, allowing for rapid adjustments to production demands while consistently maintaining the highest quality standards. Boost overall equipment effectiveness and responsiveness to market changes without compromising on excellence.



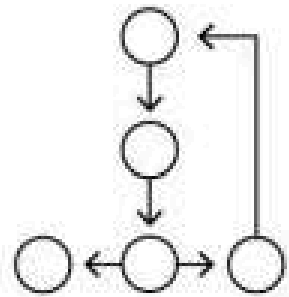
AFTERMARKET GROWTH

Unlock significant revenue streams by leveraging in-field product data for proactive service delivery and **upsell identification**. This data-driven strategy allows you to anticipate customer needs, offer timely maintenance, and identify opportunities for enhanced services or upgrades, potentially doubling your aftermarket margin potential and increasing customer loyalty.

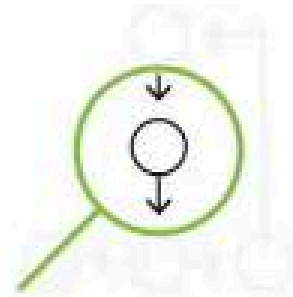
Business Value Across the Manufacturing Value Chain

A two-pronged approach for preparing for a future with agentic AI

1. Develop an agentic AI road map focused on maximizing value



Create an enterprisewide inventory of current business processes



Identify where AI agents could be embedded to generate the greatest value

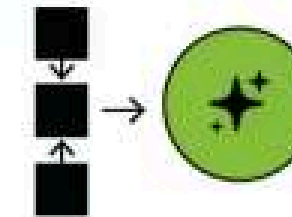


Prioritize agentic AI opportunities that offer maximum return on investment

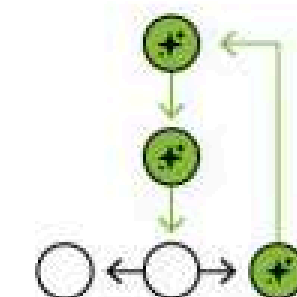
2. Implement the agentic AI road map, beginning with the highest-priority solutions



Build the technical platform and enablers



Establish governance and workforce change management mechanisms that will support larger-scale transformation



Implement pilots with prioritized opportunities and stakeholders

The Competitive Imperative of Agentic AI

EARLY ADVANTAGES

Embracing Agentic AI early is not just an option; it's a strategic necessity for securing market leadership. Early adopters will gain a significant **first-mover advantage**, setting new industry standards while competitors are still on the sidelines. This proactive approach leads to:

- **Superior Operational Excellence:** Streamlining processes and reducing costs far beyond current capabilities.
 - **Enhanced Customer Experience:** Delivering personalized, predictive, and proactive service that builds loyalty.
 - **Strengthened Supply Chain Resilience:** Anticipating disruptions and optimizing logistics with unparalleled precision.
- Furthermore, early adoption unlocks entirely new, high-margin revenue streams, particularly in aftermarket services, transforming cost centers into profit generators. Seizing this opportunity now ensures a defensible competitive moat for the future.

RISK OF INACTION

Conversely, inaction or delayed adoption of Agentic AI poses a substantial and escalating threat to a company's market position and long-term viability. The consequences of falling behind are severe and multifaceted:

1. **Erosion of Competitive Edge:** Companies that wait will find themselves struggling to catch up to new industry benchmarks for efficiency and innovation. What is a competitive differentiator today will quickly become '**table stakes**' for basic market participation tomorrow. The gap between leaders and laggards will widen exponentially, making it increasingly difficult to close.
2. **Diminished Customer Loyalty:** As competitors leverage AI to elevate service standards, customer expectations will shift permanently. Failing to meet these new, higher expectations will lead to customer churn and a loss of **competitive ground** on what is often a primary brand differentiator.
3. **Strategic Disadvantage:** The most significant risk is being relegated to a position of permanent strategic disadvantage. As Agentic AI integrates deeply into value chains, those without it will face higher costs, lower agility, and an inability to compete on data-driven insights. This isn't just about losing market share; it's about risking relevance in a transformed industrial landscape. The cost of catching up later will far exceed the investment of leading now.

Deloitte's Trustworthy AI framework to protect your AI value & investments

Applying Deloitte's 7-part Trustworthy AI Framework is an effective first step in diagnosing the ethical health of AI products while maintaining customer privacy, staying secure and abiding by relevant regulations.

SAFE & SECURE

AI systems can be protected from risks (including Cyber) that may cause physical and/or digital harm

ROBUST & RELIABLE

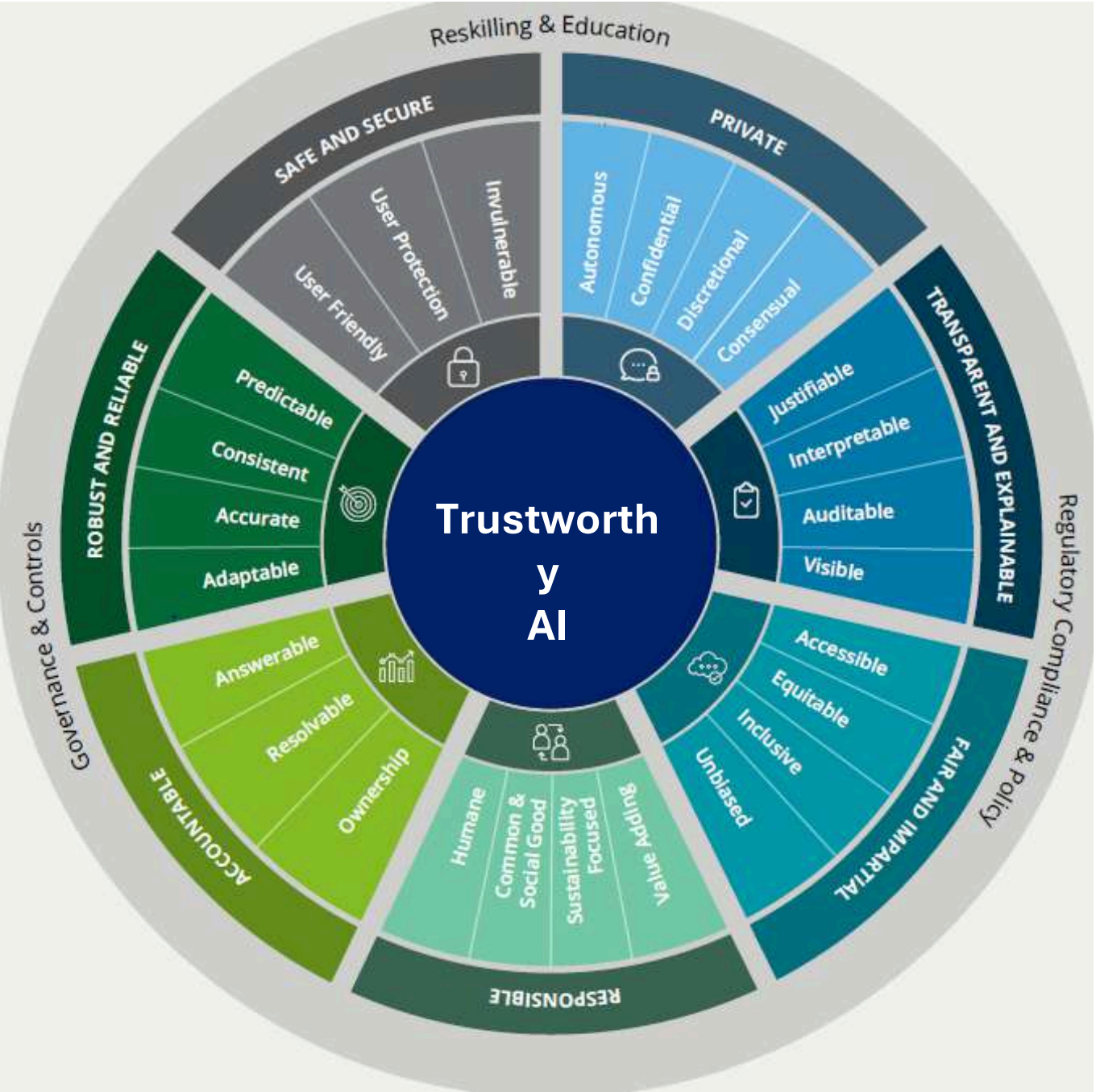
AI systems can learn from humans and other systems and produce consistent and reliable outputs

ACCOUNTABLE

Policies are in place to determine who is held responsible for the output of AI system decisions

RESPONSIBLE

The technology is created and operated in a socially responsible manner.



PRIVATE

Consumer privacy is respected, and customer data is not used beyond its intended and stated use; consumers are able to opt in/out of sharing their data

TRANSPARENT & EXPLAINABLE

All participants are able to understand how their data is being used and how AI systems make decisions; algorithms, attributes, and correlations are open to inspection

AI applications include internal and external checks to help ensure equitable application across all participants

Visit our **AI Assurance** booth (Nr 2) to learn more about **Deloitte's Trustworthy AI framework**

AI value realization is more than technology alone



Visit our **AI Toolbox** booth (Nr 1) to learn about **our tools and services** to support you in all the different dimensions of rolling out AI in your organization

THANK YOU!

Let's stay connected

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Enterprise architect

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